

Bài thực hành số 9

Bài 1:

a) Bộ lọc thông thấp

```
clear; close all; clc;

fs = 1000;
fc = 410;
N = 4;
Rp = 6;

Wn = fc/(fs/2);
[b, a] = butter(N, Wn, 'low');
[b1, a1] = cheby1(N, Rp, Wn, 'low');
t = 0: 1/fs: 0.5;
y = 2*sin(2*pi*230*t + pi/4) + cos(2*pi*400*t + 2*pi/3);

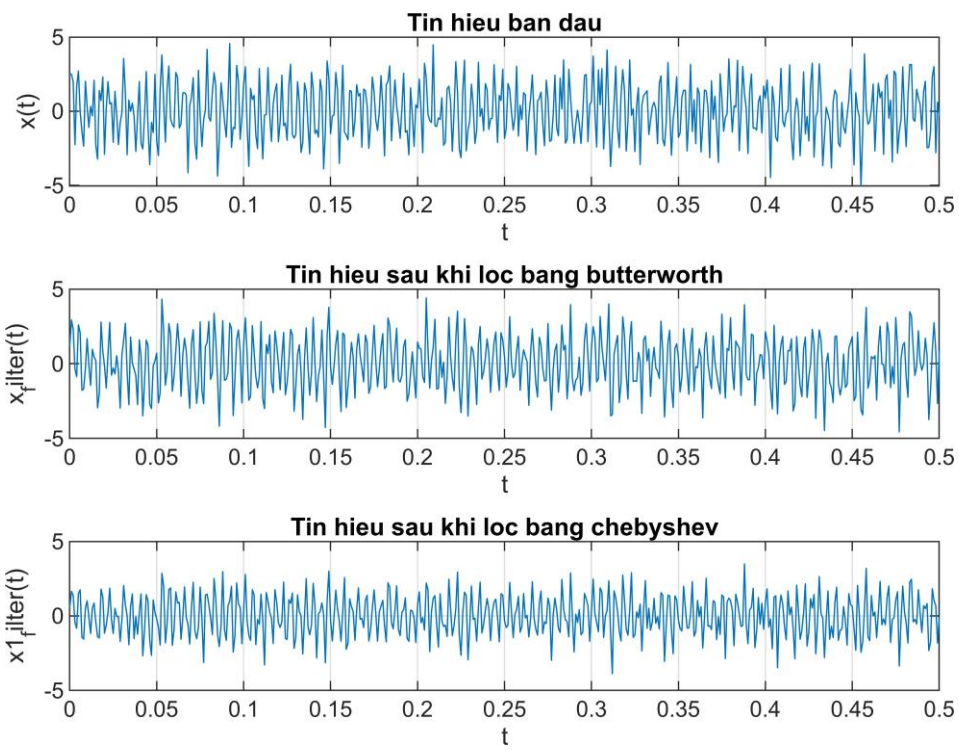
x = y + randn(size(t));

x_filter = filter(b, a, x);
x1_filter = filter(b1, a1, x);

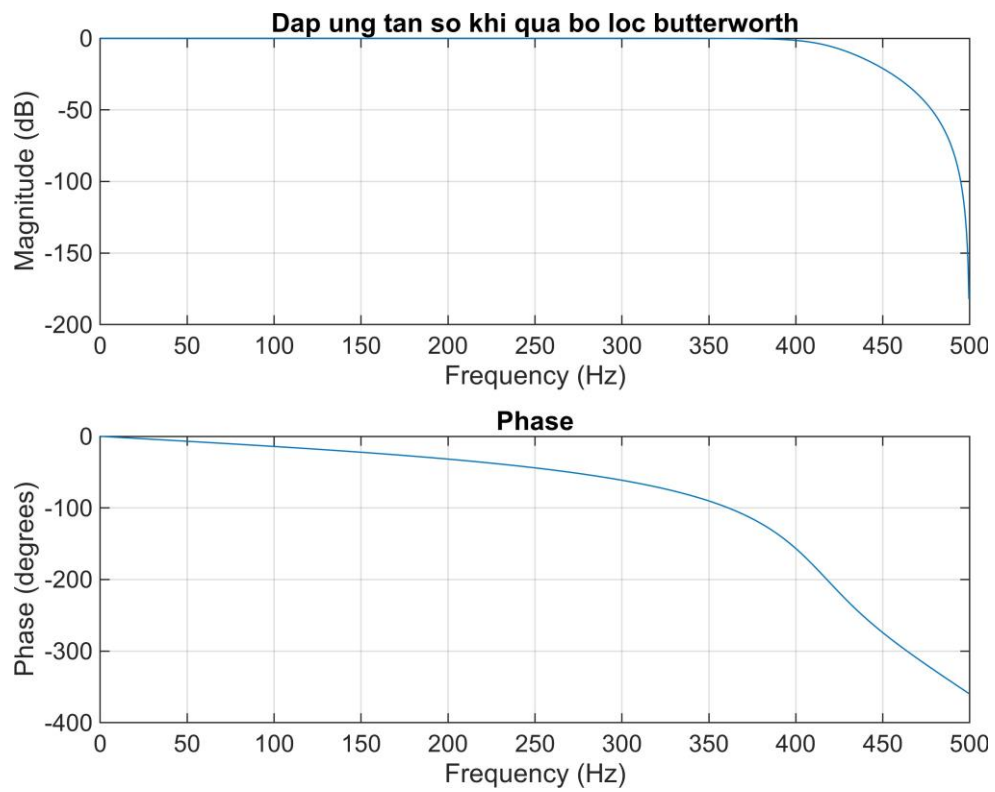
figure;
subplot(3,1,1);
plot(t, x);
xlabel("t");
ylabel("x(t)");
title("Tin hieu ban dau");
grid on;

subplot(3,1,2);
plot(t, x_filter);
xlabel("t");
ylabel("x_filter(t)");
title("Tin hieu sau khi loc bang butterworth");
grid on;

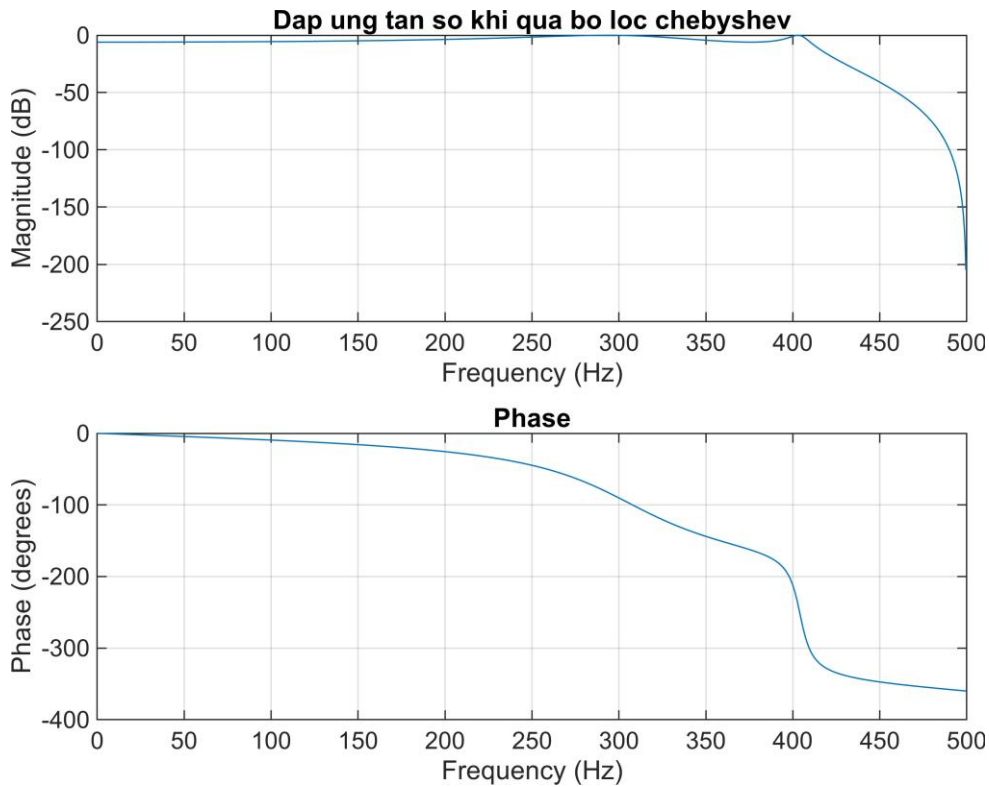
subplot(3,1,3);
plot(t, x1_filter);
xlabel("t");
ylabel("x1_filter(t)");
title("Tin hieu sau khi loc bang chebyshev");
grid on;
```



```
figure;
freqz(b, a, 1024, fs);
title("Dap ung tan so khi qua bo loc butterworth");
grid on;
```



```
figure;  
freqz(b1, a1, 1024, fs);  
title("Dap ung tan so khi qua bo loc chebyshev");  
grid on;
```



```

Nfft = length(x);
f = linspace(0, fs/2, Nfft/2);
X = fft(x, Nfft)/Nfft;
X_filter = fft(x_filter, Nfft)/Nfft;
X1_filter = fft(x1_filter, Nfft)/Nfft;

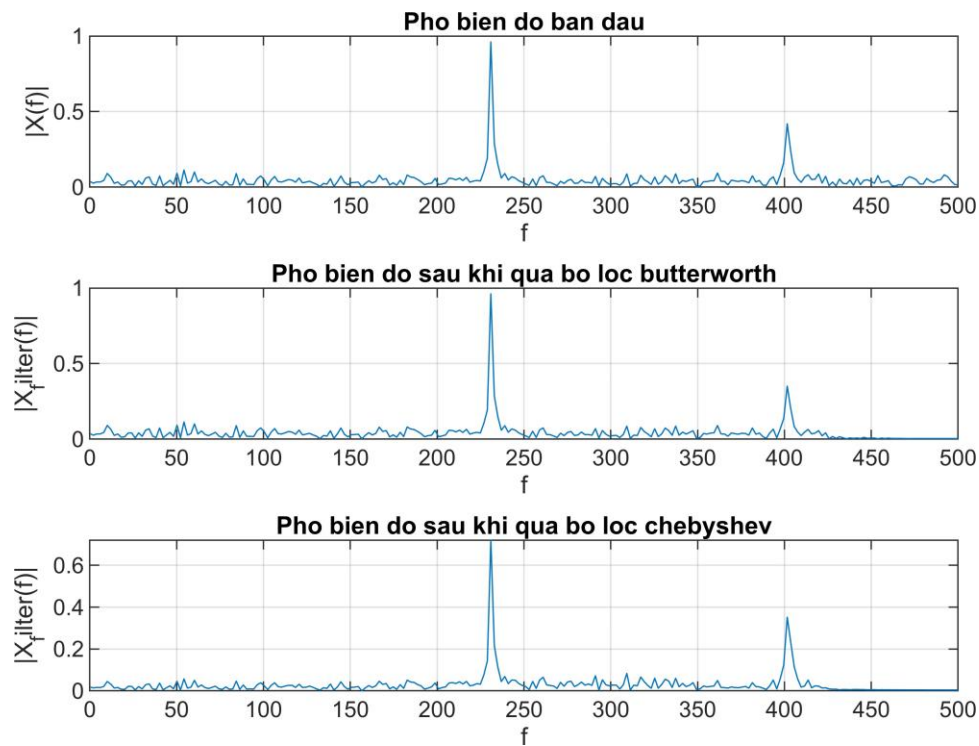
figure;
subplot(3,1,1);
plot(f, abs(X(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X(f)|");
title("Pho bien do ban dau");
grid on;

subplot(3,1,2);
plot(f, abs(X_filter(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X_filter(f)|");
title("Pho bien do sau khi qua bo loc butterworth");
grid on;

subplot(3,1,3);
plot(f, abs(X1_filter(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X1_filter(f)|");

```

```
title("Pho bien do sau khi qua bo loc chebyshev");
grid on;
```



b) Bộ lọc thông cao

```
clear; close all; clc;

fs = 1000;
fc = 200;
N = 4;
Rp = 6;

Wn = fc/(fs/2);
[b, a] = butter(N, Wn, 'high');
[b1, a1] = cheby1(N, Rp, Wn, 'high');
t = 0: 1/fs: 0.5;
y = 2*sin(2*pi*230*t + pi/4) + cos(2*pi*400*t + 2*pi/3);

x = y + randn(size(t));

x_filter = filter(b, a, x);
x1_filter = filter(b1, a1, x);

figure;
subplot(3,1,1);
plot(t, x);
```

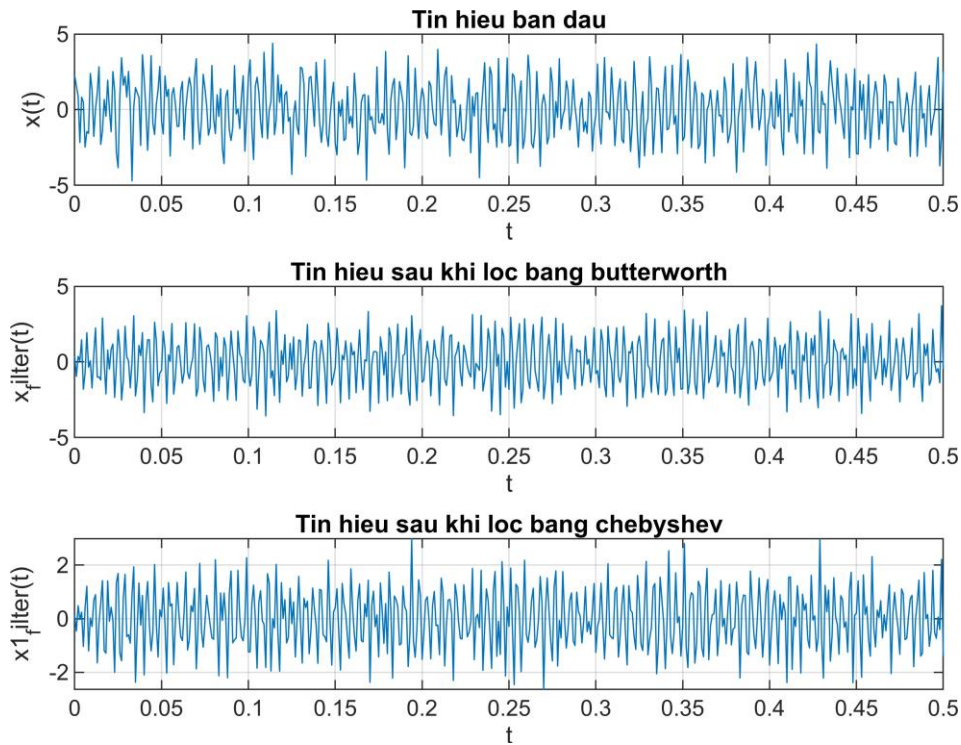
```

xlabel("t");
ylabel("x(t)");
title("Tin hieu ban dau");
grid on;

subplot(3,1,2);
plot(t, x_filter);
xlabel("t");
ylabel("x_filter(t)");
title("Tin hieu sau khi loc bang butterworth");
grid on;

subplot(3,1,3);
plot(t, x1_filter);
xlabel("t");
ylabel("x1_filter(t)");
title("Tin hieu sau khi loc bang chebyshev");
grid on;

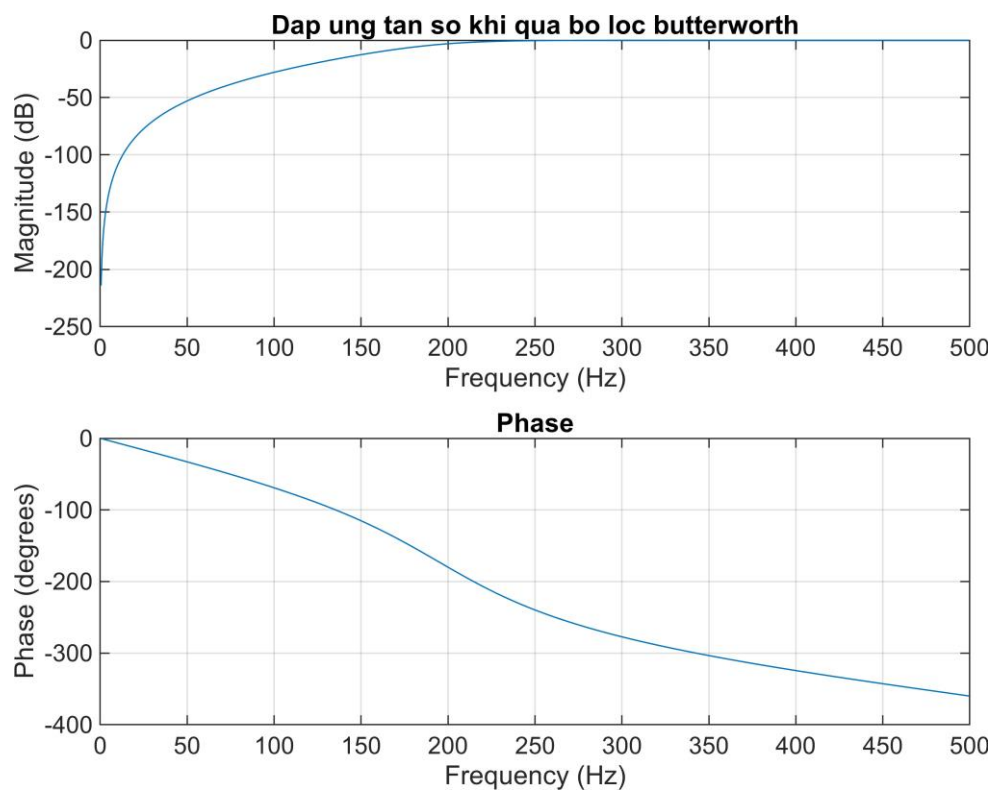
```



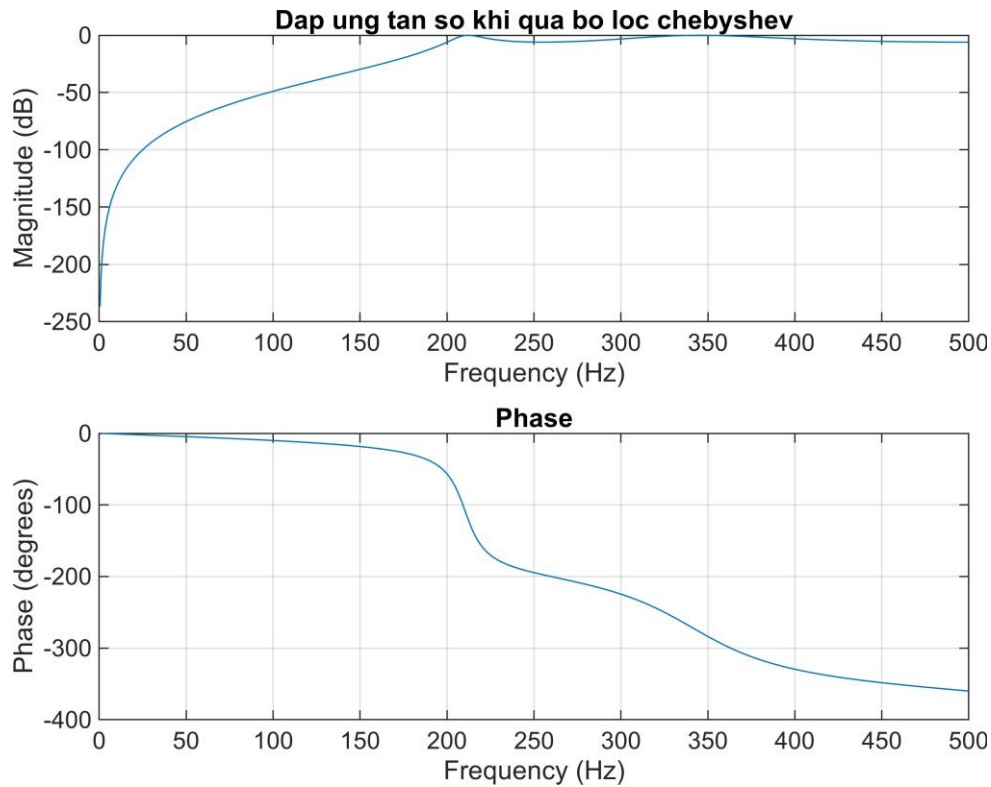
```

figure;
freqz(b, a, 1024, fs);
title("Dap ung tan so khi qua bo loc butterworth");
grid on;

```



```
figure;  
freqz(b1, a1, 1024, fs);  
title("Dap ung tan so khi qua bo loc chebyshev");  
grid on;
```



```

Nfft = length(x);
f = linspace(0, fs/2, Nfft/2);
X = fft(x, Nfft)/Nfft;
X_filter = fft(x_filter, Nfft)/Nfft;
X1_filter = fft(x1_filter, Nfft)/Nfft;

figure;
subplot(3,1,1);
plot(f, abs(X(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X(f)|");
title("Pho bien do ban dau");
grid on;

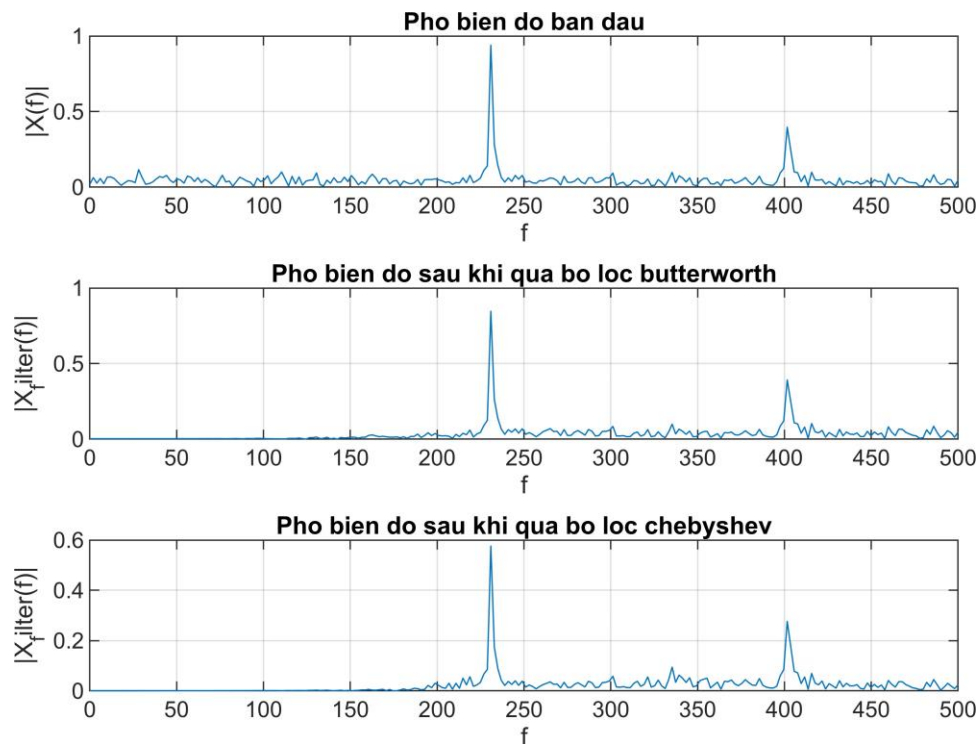
subplot(3,1,2);
plot(f, abs(X_filter(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X_filter(f)|");
title("Pho bien do sau khi qua bo loc butterworth");
grid on;

subplot(3,1,3);
plot(f, abs(X1_filter(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X1_filter(f)|");

```



```
title("Pho bien do sau khi qua bo loc chebyshev");
grid on;
```



c) Bộ lọc thông dải

```
clear; close all; clc;

fs = 1000;
f1 = 200;
f2 = 450;
N = 4;
Rp = 6;

Wn = [f1 f2]/(fs/2);
[b, a] = butter(N, Wn, 'bandpass');
[b1, a1] = cheby1(N, Rp, Wn, 'bandpass');
t = 0: 1/fs: 0.5;
y = 2*sin(2*pi*230*t + pi/4) + cos(2*pi*400*t + 2*pi/3);

x = y + randn(size(t));

x_filter = filter(b, a, x);
x1_filter = filter(b1, a1, x);

figure;
subplot(3,1,1);
```

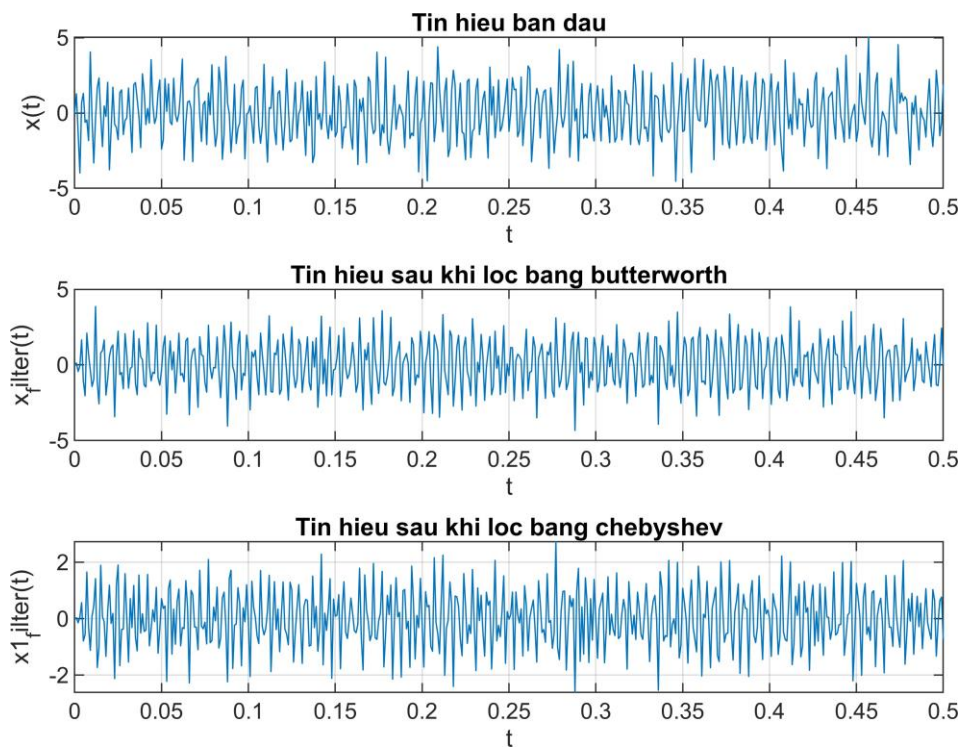
```

plot(t, x);
xlabel("t");
ylabel("x(t)");
title("Tin hieu ban dau");
grid on;

subplot(3,1,2);
plot(t, x_filter);
xlabel("t");
ylabel("x_filter(t)");
title("Tin hieu sau khi loc bang butterworth");
grid on;

subplot(3,1,3);
plot(t, x1_filter);
xlabel("t");
ylabel("x1_filter(t)");
title("Tin hieu sau khi loc bang chebyshev");
grid on;

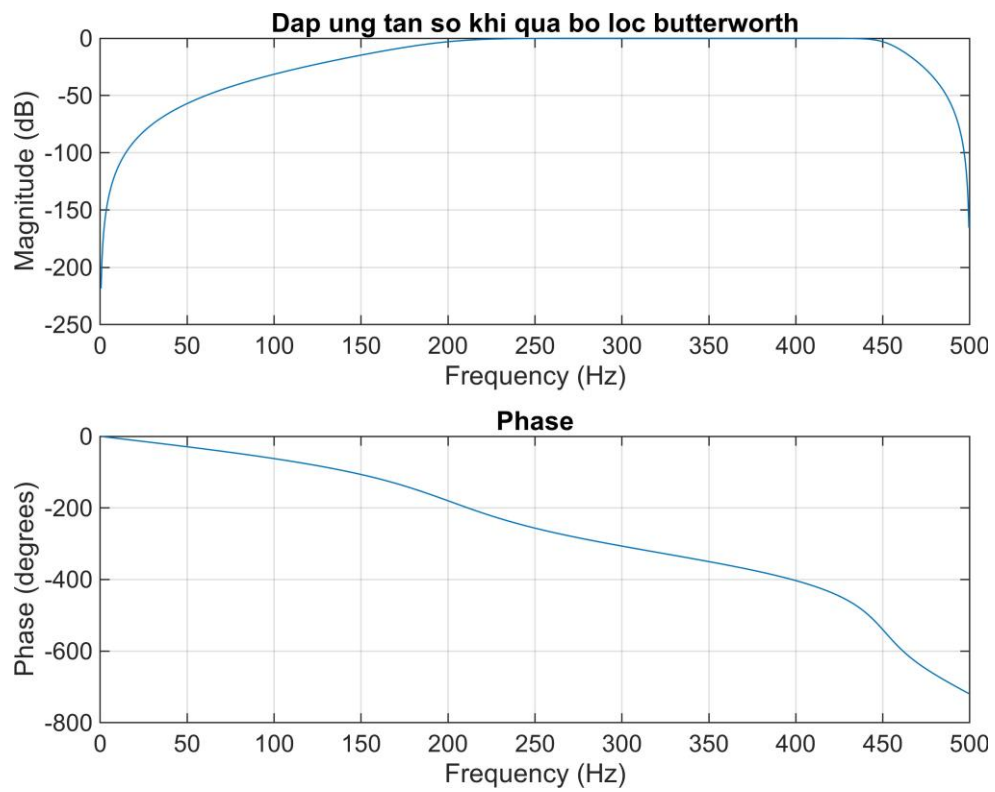
```



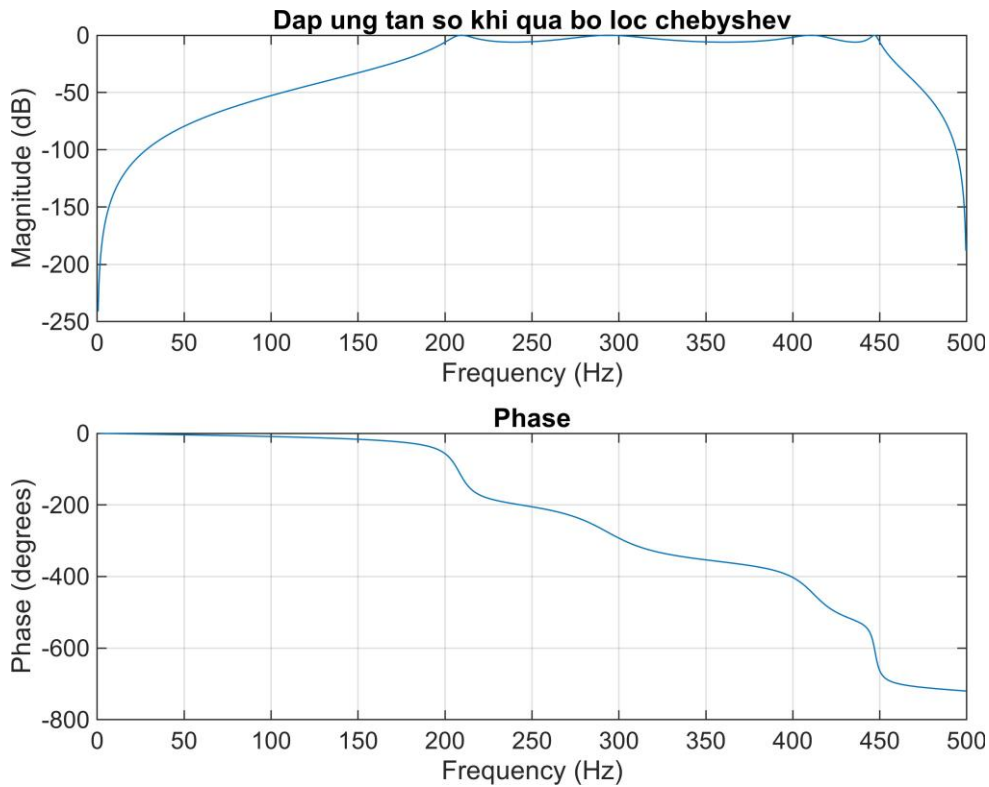
```

figure;
freqz(b, a, 1024, fs);
title("Dap ung tan so khi qua bo loc butterworth");
grid on;

```



```
figure;  
freqz(b1, a1, 1024, fs);  
title("Dap ung tan so khi qua bo loc chebyshev");  
grid on;
```



```

Nfft = length(x);
f = linspace(0, fs/2, Nfft/2);
X = fft(x, Nfft)/Nfft;
X_filter = fft(x_filter, Nfft)/Nfft;
X1_filter = fft(x1_filter, Nfft)/Nfft;

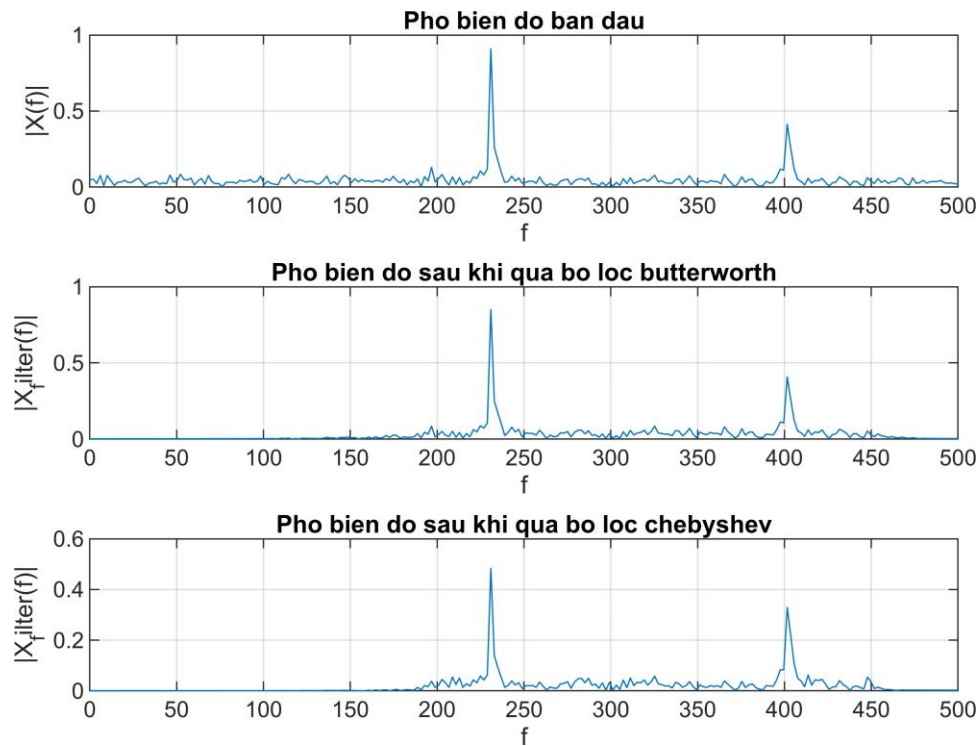
figure;
subplot(3,1,1);
plot(f, abs(X(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X(f)|");
title("Pho bien do ban dau");
grid on;

subplot(3,1,2);
plot(f, abs(X_filter(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X_filter(f)|");
title("Pho bien do sau khi qua bo loc butterworth");
grid on;

subplot(3,1,3);
plot(f, abs(X1_filter(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X1_filter(f)|");

```

```
title("Pho bien do sau khi qua bo loc chebyshev");
grid on;
```



d) Bộ lọc triệt dài

```
clear; close all; clc;

fs = 1000;
f1 = 250;
f2 = 350;
N = 4;
Rp = 6;

Wn = [f1 f2]/(fs/2);
[b, a] = butter(N, Wn, 'stop');
[b1, a1] = cheby1(N, Rp, Wn, 'stop');
t = 0: 1/fs: 0.5;
y = 2*sin(2*pi*230*t + pi/4) + cos(2*pi*400*t + 2*pi/3);

x = y + randn(size(t));

x_filter = filter(b, a, x);
x1_filter = filter(b1, a1, x);

figure;
subplot(3,1,1);
```

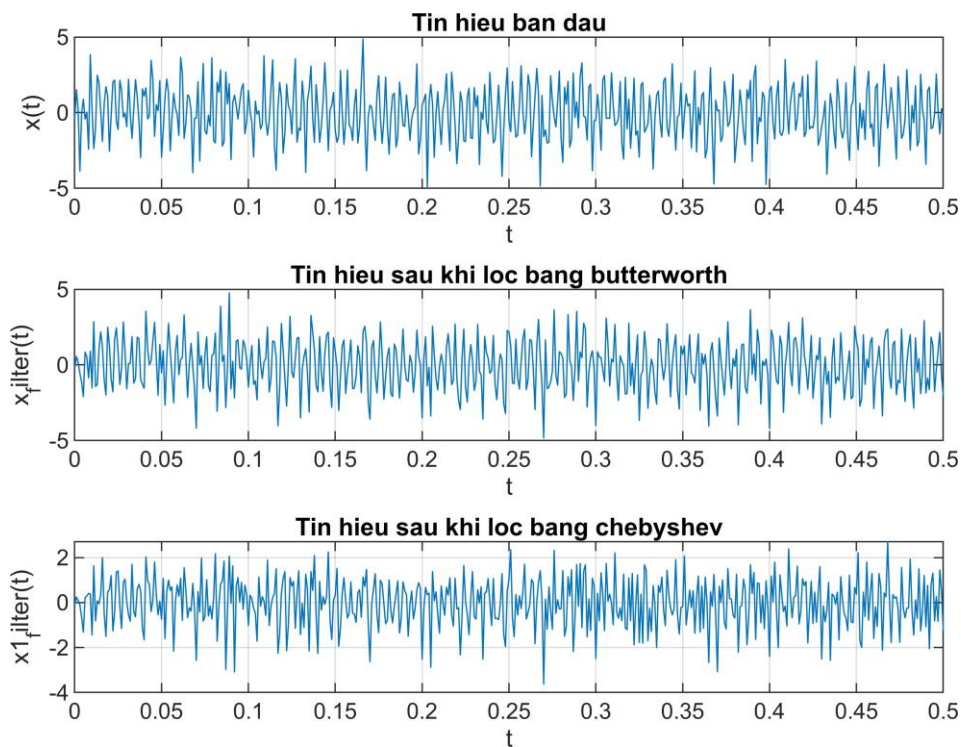
```

plot(t, x);
xlabel("t");
ylabel("x(t)");
title("Tin hieu ban dau");
grid on;

subplot(3,1,2);
plot(t, x_filter);
xlabel("t");
ylabel("x_filter(t)");
title("Tin hieu sau khi loc bang butterworth");
grid on;

subplot(3,1,3);
plot(t, x1_filter);
xlabel("t");
ylabel("x1_filter(t)");
title("Tin hieu sau khi loc bang chebyshev");
grid on;

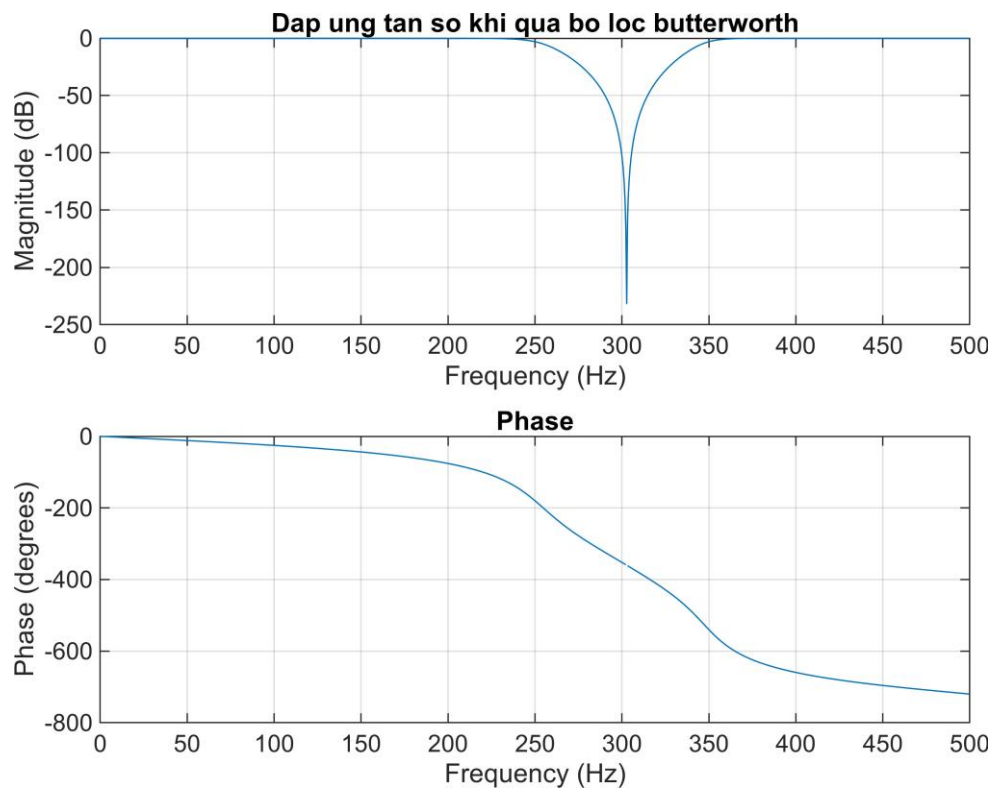
```



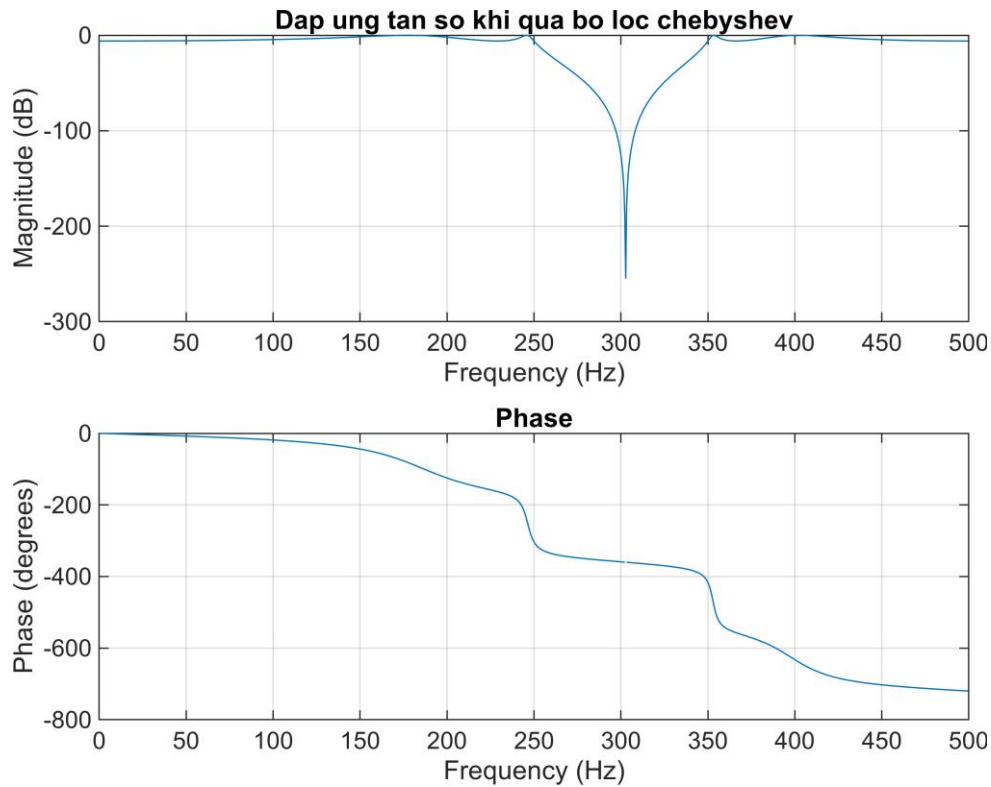
```

figure;
freqz(b, a, 1024, fs);
title("Dap ung tan so khi qua bo loc butterworth");
grid on;

```



```
figure;  
freqz(b1, a1, 1024, fs);  
title("Dap ung tan so khi qua bo loc chebyshev");  
grid on;
```



```

Nfft = length(x);
f = linspace(0, fs/2, Nfft/2);
X = fft(x, Nfft)/Nfft;
X_filter = fft(x_filter, Nfft)/Nfft;
X1_filter = fft(x1_filter, Nfft)/Nfft;

figure;
subplot(3,1,1);
plot(f, abs(X(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X(f)|");
title("Pho bien do ban dau");
grid on;

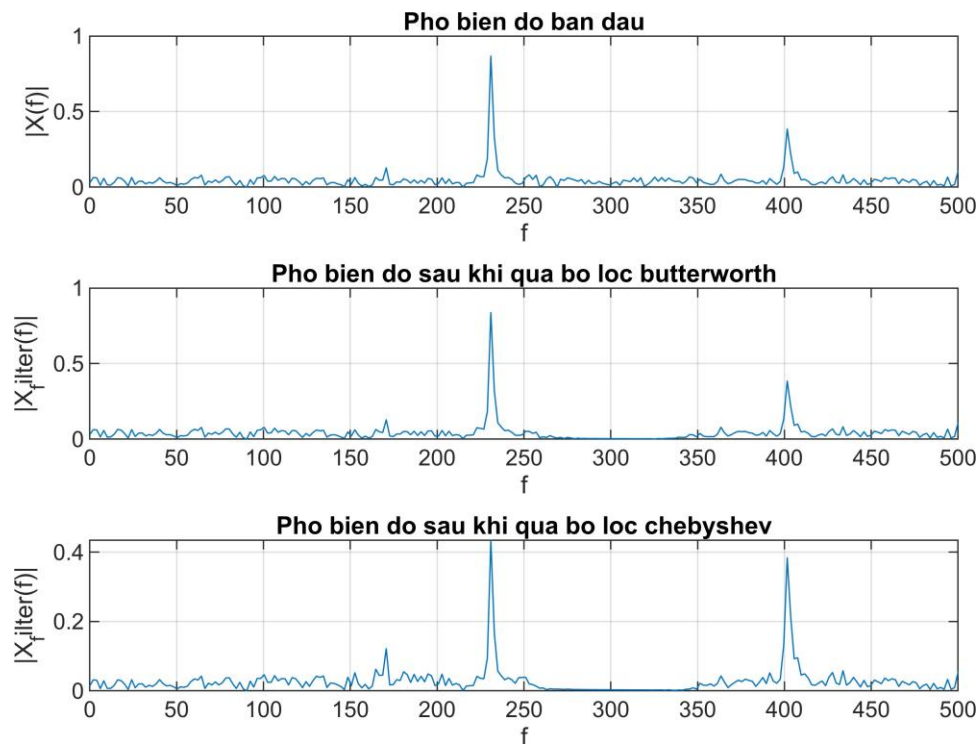
subplot(3,1,2);
plot(f, abs(X_filter(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X_filter(f)|");
title("Pho bien do sau khi qua bo loc butterworth");
grid on;

subplot(3,1,3);
plot(f, abs(X1_filter(1:floor(Nfft/2))));
xlabel("f");
ylabel("|X1_filter(f)|");

```



```
title("Pho bien do sau khi qua bo loc chebyshev");
grid on;
```



Bài 2:

```
clear; close all; clc

fs = 8192;
f1 = 210;
f2 = 390;
N = 4;

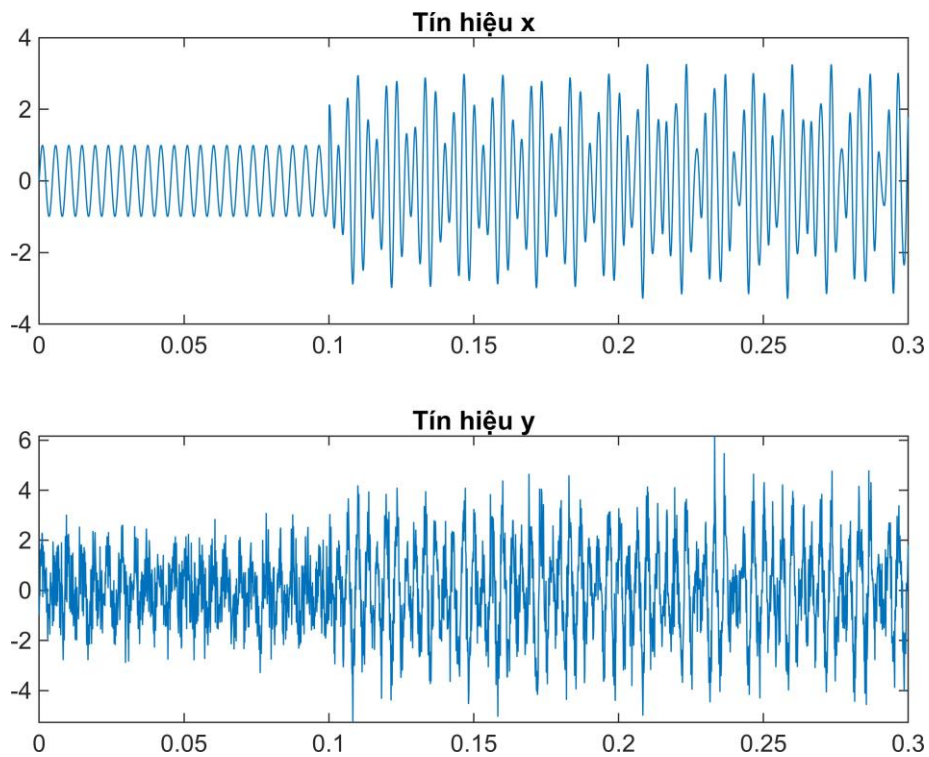
Wn = [f1 f2]/(fs/2);
[b,a] = butter(N, Wn, "stop");

t = 0: 1/fs: 0.3;
x1 = (t>0 & t<=0.3) .* (sin(2*pi*220*t));
x2 = (t>=0.1 & t<=0.3) .* (2*cos(2*pi*300*t));
x3 = (t>=0.2 & t<=0.3) .* (0.5*sin(2*pi*440*t));

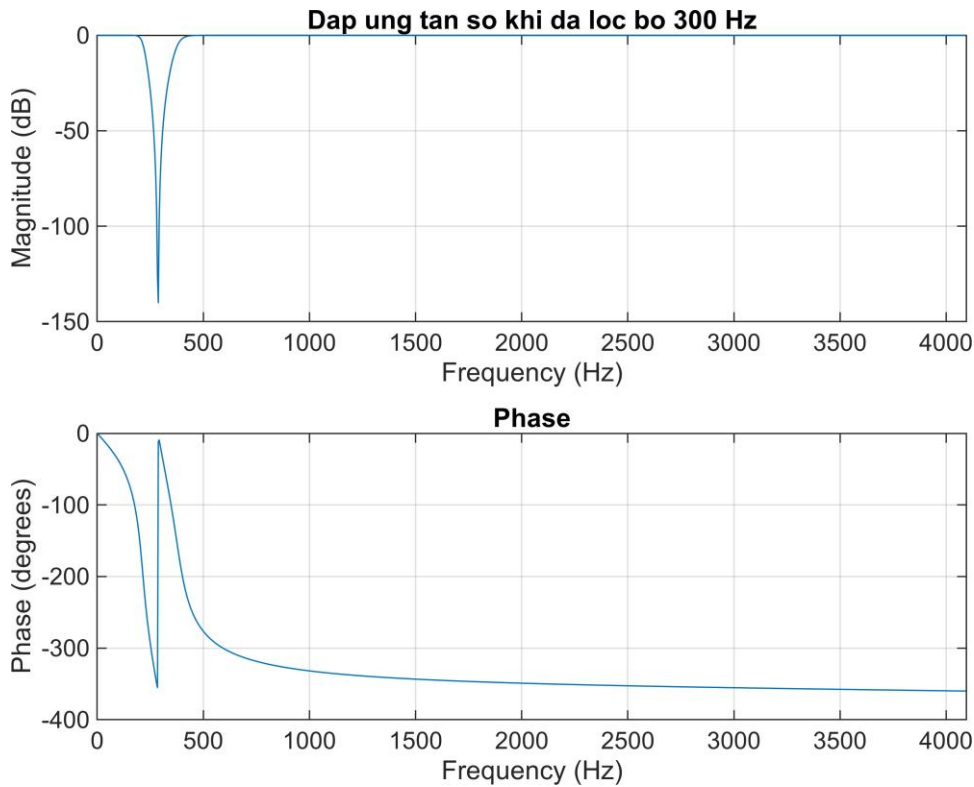
x = x1+x2+x3;
noise = randn(size(t));
y = x + noise;

figure;
subplot(2,1,1); plot(t, x); title("Tín hiệu x");
```

```
subplot(2,1,2); plot(t, y); title("Tín hiệu y");
```



```
figure;  
freqz(b, a, 1024, fs);  
title("Đáp ứng tần số khi đã lọc bỏ 300 Hz");  
grid on;
```



```

y_filtered = filter(b, a, y);

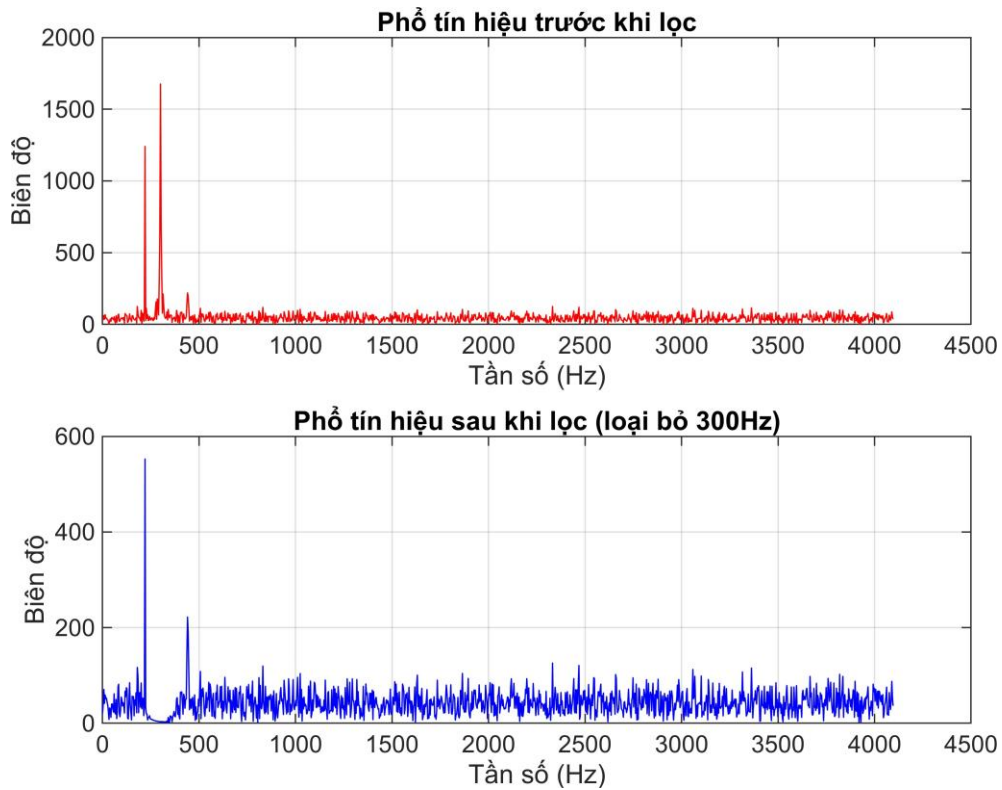
Nfft = length(y);
f = linspace(0, fs/2, Nfft/2);

Y = fft(y, Nfft);
Y_filtered = fft(y_filtered, Nfft);

figure;
subplot(2,1,1);
plot(f, abs(Y(1:Nfft/2)), 'r');
xlabel('Tần số (Hz)');
ylabel('Biên độ');
title('Phổ tín hiệu trước khi lọc');
grid on;

subplot(2,1,2);
plot(f, abs(Y_filtered(1:Nfft/2)), 'b');
xlabel('Tần số (Hz)');
ylabel('Biên độ');
title('Phổ tín hiệu sau khi lọc (loại bỏ 300Hz)');
grid on;

```



Bài 3:

```
clc; clear; close all;
load('ECGdata.mat')

y0 = ECG0(:);
Nfft = length(y0);
t = (0: Nfft-1)*1/fs0;

f = linspace(0, fs0/2, Nfft/2);
ECG_fft = fft(y0, Nfft)/Nfft;

%ve tin hieu ECG khi chua loc nhieu
figure;
subplot(3,1,1);
plot(t, y0);
xlabel("t");
ylabel("y0(t)");
title("Tin hieu ECG ban dau");
grid on;

%ve pho ban dau
subplot(3,2,3);
plot(f, 20*log(abs(ECG_fft(1:floor(Nfft/2)))));
xlabel("f(Hz)");
```

```

ylabel("|ECG_fft| (dB)");
title("Pho tin hieu ban dau");
grid on;

%loai bo tin hieu co tan so 50 Hz
[b, a] = butter(4, [49.5 50.5]/(fs0/2), 'stop');

y = filter(b,a, y0);
Y_fft = fft(y, Nfft)/Nfft;
subplot(3,2,4);
plot(f, 20*log(abs(Y_fft(1:floor(Nfft/2)))));
xlabel("f(Hz)");
ylabel("|Y_fft|(dB)");
title("Pho tin hieu khi da loc tan so 50 Hz");
grid on;

%giu lai tin hieu co tan so 0.5 Hz den 45 Hz
[b1, a1] = cheby1(4, 10, [0.5 45]/(fs0/2), 'bandpass');

y1 = filter(b1,a1, y);

Y1_fft = fft(y1, Nfft)/Nfft;
subplot(3,2,5);
plot(f, 20*log(abs(Y1_fft(1:floor(Nfft/2)))));
xlabel("f(Hz)");
ylabel("|Y1_fft| (dB)");
title("Pho tin hieu khi da loc nhieu xong");
grid on;

%ve tin hieu ECG0 khi da loc nhieu xong
subplot(3,2,6);
plot(t, y1);
xlabel("t");
ylabel("y1(t)");
title("Tin hieu ECG0 da loc nhieu");
grid on;

```

